

Econ 2 - Lecture 14 - 3/2/26

Lecture Quiz #8 Released Wednesday, 3/4

↳ Due Monday, 3/9

Note: Final Exam on Wednesday, 3/18 (40 MC Questions)

Today: Finish Debt (Chapter 5.2), Monetary Policy (Chapter 6)

Next Week: Aggregate Demand, Aggregate Supply (Chapter 7)

Last Class:

Total Debt = Public Debt + Intragovernmental Holdings

↑
Fact #1

Paying Back Debt → Roll over indefinitely

↑
Fact #2

$$\text{Minimum Tax Rate} = \underbrace{\text{Interest Rate}}_{\text{Monetary Policy}} \times \underbrace{\frac{\text{Public Debt}}{\text{Nom. GDP}}}_{\frac{\text{Debt}}{\text{GDP}} \text{ Ratio}} \times 100$$

Fact #3: Relevant debt statistic = $\frac{\text{Debt}}{\text{GDP}}$ ratio

As $\frac{\text{Debt}}{\text{GDP}}$ ratio ↑, min. tax rate ↑

Low MPC

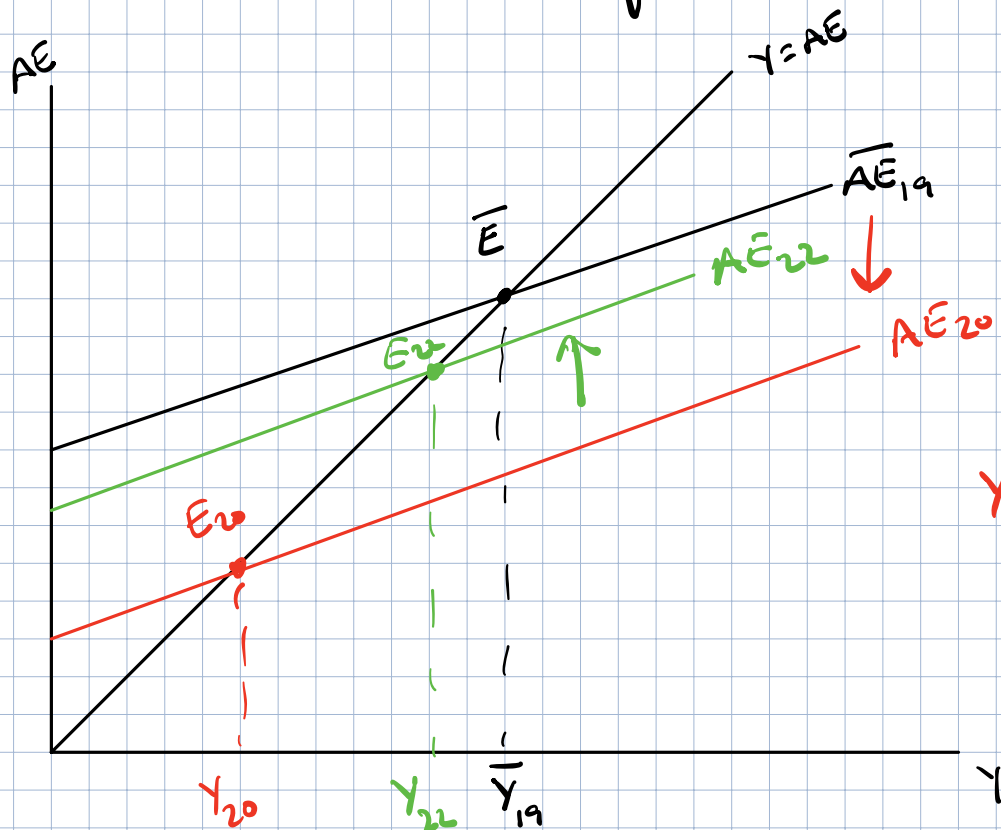
Fact #4: Interest payments are not gov't purchases

No production from interest payments

~1/3 of debt owned by foreign investors

Safety/stability in owning US Dollars/Treasury

Summarize: Short-Run Equilibrium + Shocks + Fiscal Policy + Debt



Start at $\bar{Y}_{19}$

1) Covid
Pandemic
 $\Rightarrow \downarrow AC, \downarrow I^P$
 $\Rightarrow \downarrow AE$

$Y_{20} < \bar{Y}$

2.) Stimulus
Package

$\Rightarrow \Delta G > 0$
 $\Delta T < 0$

Increasing
Debt by "STr.l"
 $\Rightarrow \uparrow AE$

By 2023 $\rightarrow$ Return to $\bar{Y} \rightarrow$ Higher
Debt Burden
Min. tax rate $\uparrow$

$\frac{\text{Debt}}{\text{N. GDP}}$ ratio?

Feed

$$\text{Min. Tax Rate} = \text{Interest Rate} \times \frac{\text{Sum } (G - T)}{\text{Prices} \times \text{Real GDP}}$$

Can we reduce minimum tax rate, but keep $Y = \bar{Y}$?

1.) Decrease Deficit: $G \downarrow$ or $T \uparrow \Rightarrow \downarrow AE$

2.) Increase Prices $\Rightarrow$ increase in Nom. GDP
 $\hookrightarrow \downarrow AE$

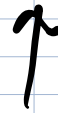
$\rightarrow$ technology booms

3.) Encourage/incentivize efficient increase
in Real GDP

Next Part of Course: Monetary Policy

Same goal as before: Get Y back to $\bar{Y}$

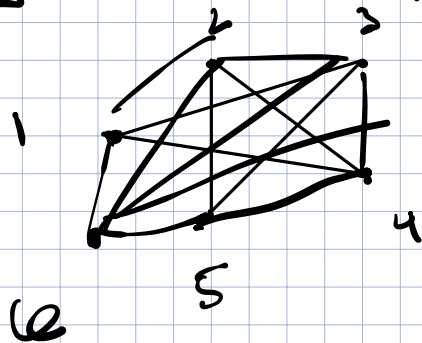
Connect monetary policy actions to $Y = AE$



Define, model, enact $\rightarrow$ consequences

Money? A world without money?

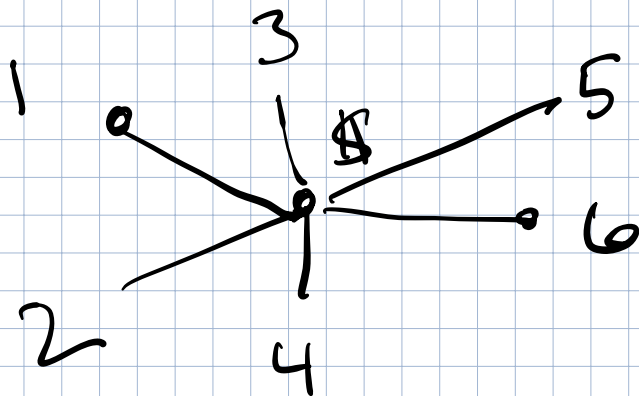
Bartering $\rightarrow$ 5 Goods, trading pairs



$$\frac{5 \times 4}{2} = 10 \text{ trading Pairs}$$

$$50 \text{ goods} \rightarrow \frac{50 \times 49}{2} = 1,225 \text{ Pairs}$$

Money: Good that is always on one side of a trade



Efficiency
of trade

3 Main Characteristics of money

1. Medium of Exchange → used by individuals & firms
2. Unit of Account → every good is valued by "money"
3. Store of Value → Long lasting, and preserve value

Commodity Money: Money with intrinsic value

Examples: Gold, Salt, Spices, silk, furs, silver,
✓ Rai Stones on Yap Island (Granite)

Only able to mine 1-2% of gold stock per year

US Dollar? Before 1970s → \$35 → 1 oz of gold

After 1972 → No backing of US Dollar

→ Fiat Currency: No intrinsic value

↓ ↳ Backed by faith in govt & usability

Norm Today

Categories of Money: Based on Liquidity

Liquidity: ability to use the form of money to buy goods/services

M1: Most liquid form of money

→ Cash & Coins ⇒ 2.3 Trillion

→ Debit ⇒ Checking Acct. ⇒ 18 Trillion

Credit Card? Short-term loan

→ Means of exchange, but not a store of value

M2: M1 + Time Deposits

→ Savings Account → At least one step between Savings acct. & lunch!

→ Money market funds

→ Certificates of Deposits (CDs)

Money Supply = M1 money only

Banks: Broker between savers & borrowers

Central Bank: Bank for banks

Federal Reserve = "The Fed" ≠ Gov't / Congress

Monetary Policy: 8 times year at Federal Open Market Committee Meetings (FOMC)

↓
FOMC ⇒ Chair, 6 Board Members,
12 Branch Presidents ⇒ 19 voices
Fewer voices than Fiscal Policy

Two Primary Goals of Fed

1.) Maximum Employment $\rightarrow Y$

2.) Price Stability

Monetary Policy: changing the M1 money supply, through open-market operations, with the goal of altering short-term interest rates

Open-market purchases:

Fed buys bonds from public,
public give bonds to Fed $\rightarrow$ Fed give money to public

Increase the money supply

Open-market sales:

Fed sells bonds to public,

Public gives money to Fed, receive bond

Decrease the money supply

Bonds in monetary policy

Example: \$100, one-year bond

In one year, bond matures, converts to \$100

	Bond	\$100
3/2/27		

Maximum Price? ~\$99.99

Price $< 100 \Rightarrow$ Earn interest

If Price = \$75 $\rightarrow$ Earn \$25

Federal Reserve target Federal Funds Rate
(FFR)

FFR: Interest banks charge each other
(overnight loans)

Lower bound of interest rates

Develop a model for M1 money $\rightarrow$ connect
to AE